UNIVERSITY OF AMSTERDAM

ASSIGNMENT 1

Exploring Real World Networks: Wikipedia "Squirrel" database

November 6, 2026

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Course:
Model-Based Decision Making

1 Introduction and Dataset Context

The assignment required acquiring and exploring an empirical network dataset of interest. Using the database of SNAP by Stanford University (Leskovec and Krevl 2014), I selected the Wikipedia Article Networks, which represents page-to-page links for a specific topic. It is part of a bigger research project, MUSAE, which is presented in (Rozemberczki, Allen, and Sarkar 2019). Nodes are represented as ID codes starting from 0, while edges correspond to the rows in the dataset, with $id1$ origin and $id2$ destination of the link between pages. The selected dataset from the repository is "Squirrel", with 5201 nodes and 198353 edges before cleaning. The dataset will be analyzed as an undirected graph.

2 Empirical Topological Analysis

Some statistical analysis has been run on the dataset and underlying network. The graph was found to be fully connected, with a single connected component. Considering it represents hyperlinks between Wikipedia pages, this is as expected. The average degree is quite high, at $\overline{d} = 76.275$, suggesting high interconnectedness. However, as shown in (Figure 1), distribution of degrees is very skewed. Low network density ($\delta = 0.014668$), alongside this skewed distribution of degrees, and negative assertativity (-0.2272) may indicate a hierarchical network structure: a small minority of articles are highly connected, linking together the majority of articles, being less-connected, possibly specialized. Connectivity, as will be explored in section 4, is indeed distributed unevenly in the network, which might make it less robust to removing important links between pages.

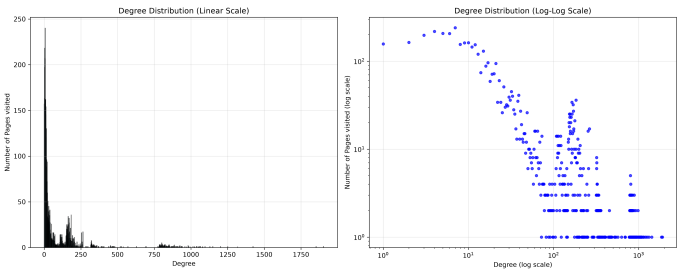


Figure 1: Degree distribution for Squirrel network

Some other measurements are representative of the network: diameter being 10 reflects exactly that the network is highly clustered, so shortest paths are quite short. Similarly, the average shortest path being 3.09 supports this. The clustering measurements of the network, moreover, suggest that interconnection tends to cluster in neighbourhoods rather than being sparse. That is explainable, in the case of "Squirrel" Wikipedia pages, by assuming articles about the animal from a scientific perspective will link to each other, while those on history, geography, domestication, etc. of the animal will presumably cluster into their own respective fields. Figure 2 presents a depiction of the network as developed in Gephi (Bastian, Heymann, and Jacomy 2009), with colours representing the clusters found in the analysis.

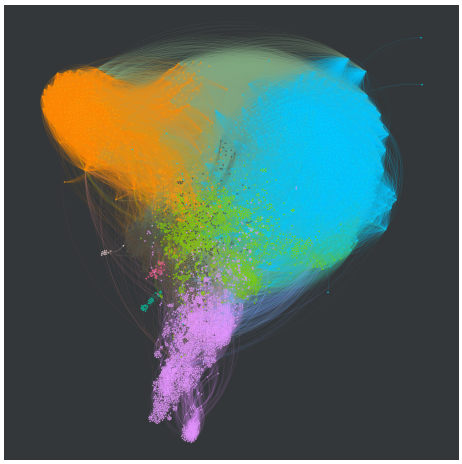


Figure 2: Depiction of network "Squirrel" Wikipedia coloured by clusters

3 Model Network Benchmarking

The network was compared with model networks from literature. For this, the following models were created with same amount of nodes and edges (simulated links) as in the empirical model "Squirrel": Erdos-Renyi, Watts-Strogatz, Barabasi-Albert, through the functions in library NetworkX (NetworkX Developer Team (n.d.)). This was done to understand which model of network shows the most similarities to the empirical network, and whether the latter follows standard patterns in its structure. As presented in Barabási 2016, the canonical models differ in their structure and dynamics: Erdos-Renyi simulates a random network model, where edges have a probability of appearing. On the contrary, the Watts-Strogatz model tends to create clusters in a so called small-world structure, with low average path lengths. Lastly, Barabasi-Albert also follows a "rich-to-the-rich" dynamic, with edges developing subsequently with higher probability where the node already has a high degree, creating a small percentage of nodes with high degree. The following Table 1 presents the results of the comparison between empirical and synthetic models, while Figure 3 presents the degree distributions comparison between models.

Table 1: Comparison between networks

Network	Nodes	Edges	Avg. Degree	Clustering	Avg. Path Length	Assortativity
Empirical	5201.0	198353.0	76.2749	0.4222	3.0906	-0.2272
Erdos-Renyi	5201.0	198353.0	76.2749	0.0146	2.3072	-0.0040
Watts-Strogatz	5201.0	197638.0	76.0000	0.6360	2.8656	-0.0011
Barabasi-Albert	5201.0	196194.0	75.4447	0.0461	2.2719	-0.0012

Some interesting inferences can be drawn from the comparison between network models. As set in their creation, they all have equal number of nodes, $N = 5201$. Edges slightly vary, due to the randomicity in construction and inherent mechanisms of each model, but all this within a 1.5% range (∼2000 out of 198353). The average degree of nodes is thus also very similar between models. As shown in Figure 3, however, its distribution is very different between models: while

Erdos-Renyi and Watts-Strotgatz present distributions centered around the average, Barabasi-Albert and the empirical network both have skewed distributions with the majority of nodes having very low degree, and a few with very high degree. The empirical model shows a longer tail than all other models, meaning some nodes are extremely important in how many edges they are part of.

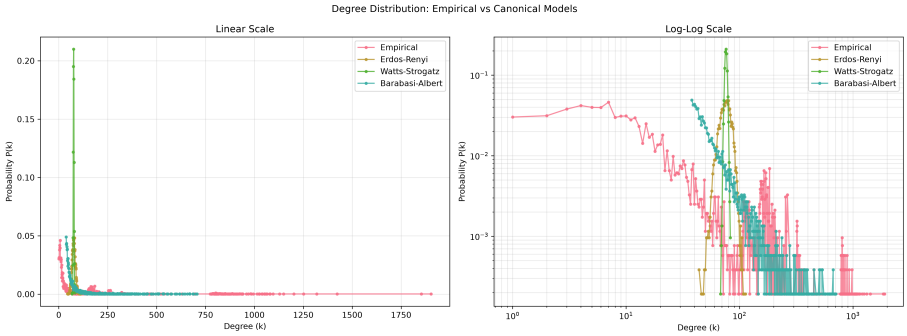


Figure 3: Comparison of Degree distribution between models

Slightly longer avg. path length (3.09) and lower assortativity (−0.23) of the empirical model compared to all simulated models shows the phenomenon mentioned in section 2: the connectivity in the network is distributed unevenly, meaning that some nodes have a central role in preserving the connectivity of the network, namely those pages on "Squirrel" that are more general and longer, thus having more links to and from them, while other, possibly specialized pages, are nodes with little edges connected to them. Indeed, assortativity (range [-1,+1]), measuring whether edges with similar degrees connect to each other more, being negative shows that nodes with higher degrees tend to connect to those with lower degree, and viceversa, differing from all simulated models where assortativity is closer to 0. This also is reflected in the average path length being slightly higher than in simulated models.

Overall, none of the simulated models seems to replicate well the empirical network analyzed, exactly because of the unique characteristic presented above, i.e. its hierarchical structure in connections. Nevertheless, it presents major differences from Erdos-Renyi, both in greater connectedness compared to it (ER is random model, thus components are not necessarily all connected), and degree distribution. The other two simulated models are both representative of some mechanics of the empirical model: the distribution of Barabasi-Albert well represents how nodes with higher degree attract even more links, just like in longer Wikipedia articles being more probably updated with even more links from and to them; moreover, the clustering being very high in the empirical model (0.4222), similarly to Watts-Strogatz (0.6360), shows characteristics of a small-world network, since clusters between pages from the same field tend to link to each other more than they link to pages on other fields connected to squirrels.

4 Centrality and node importance

Centrality analysis has been conducted using betweenness, closeness, degree, and eigenvector centrality. The results consistently identify pageID=4346 as the most central node across all four measures. It obtains a degree centrality of 0.366, betweenness centrality of 0.131, closeness centrality of 0.520, and eigenvector centrality of 0.071. This suggests that the page occupies a particularly prominent position in the network. It is possible that this page corresponds to the main Wikipedia article for "Squirrel", which would explain its centrality, since many articles related to the topic might link to it. However, because neither the database nor the project provides the dictionary between page IDs and their corresponding Wikipedia articles, this interpretation cannot be verified.

Moreover, the results reveal considerable overlap between the nodes ranked highest by the different centrality measures. In particular, pageID=5112 ranks second across all four measures, with degree, betweenness, closeness, and eigenvector centralities of 0.356, 0.093, 0.515, and 0.070,

respectively. Other pages, such as 4365, 4903, and 4303, also appear among the top five for multiple measures. This consistency suggests that the importance of these pages is not limited to a single structural property.

The different measures capture complementary aspects of node importance. Degree centrality measures the number of direct connections a page has; betweenness centrality measures how frequently it lies on shortest paths between other pages; closeness centrality captures its average distance from the rest of the network; and eigenvector centrality accounts for both the number and the centrality of its neighbours.

Overall, the results suggest that the network's structure is strongly influenced by a relatively small number of prominent pages. For instance, the highest degree centrality score is approximately 6.4 times the score of the tenth-ranked page (0.366 compared to 0.208). Similarly, the highest betweenness score is more than three times that of the third-ranked page (0.131 compared to 0.040). Such pages may play an important role in information diffusion by providing short paths between different parts of the network and connecting users to a large number of other articles. However, centrality alone does not establish how users actually navigate Wikipedia, since the analysis captures the hyperlink structure rather than observed browsing behaviour.

5 System Dynamics and Policy Implications

The results presented in the sections above give a clear picture of the structure and mechanics of the network explored. Firstly, high clustering within nodes (compared to classical simulated models) indicate a small-world behaviour of the network, being split in different highly interconnected groups. As mentioned before, this might suggest that Wikipedia articles about squirrels from a common perspective (science, taxonomy, geographic distribution, etc.) tend to reference each other, thus showing these clusters. Secondly, the low network density, alongside a skewed distribution of degrees and negative assertivity all point to a hierarchical structure of the network (scale-free) with an uneven distribution of connectivity between nodes. That is to say, there are some central highly connected nodes (e.g. id=4346, possibly the entry "Squirrel") that are connected to a high number of low degree nodes (possibly specialized pages on specific topics). This all reflects the assumptions of how Wikipedia works: few general articles hold the majority of hyperlinks to more specific articles, that are generally less connected to other pages than their "parent" page (e.g. the page regarding ethimology of the word squirrel). The unique structure of the network has implications for the robustness of the model: exactly because some articles are disproportionately important in maintaining connectivity in the network, removing highly ranked articles (in centrality measures) removes a larger amount of paths compared to doing the same in simulated classical models. Robustness to node removal, both random and according to highest ranking, has been tested and is presented in Figure 4. As shown, indeed, a removal of 50% of the nodes leads to the loss of 80% of the network size, when done to the nodes with highest degree, while to 50% of it assuming random removal.

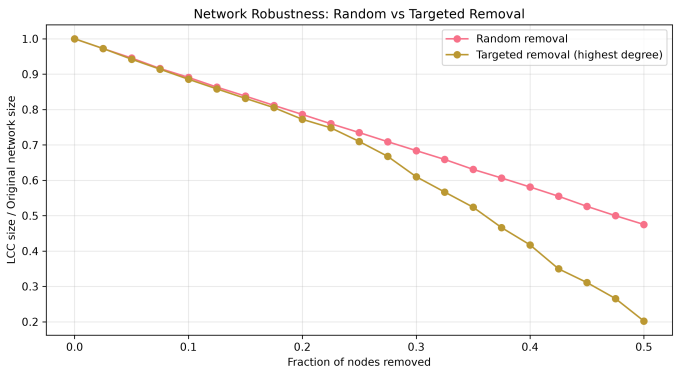


Figure 4: Robustness of network to node removal

The consequence is that, as introduced before, the network is susceptible to large losses if specific nodes are targeted. That is, however, inevitable considering that Wikipedia's structure itself is that of an encyclopedia where more common articles/pages are supposed to hold the majority of hyperlinks to more specific pages, and vice versa. It is thus neither immediate nor necessary to fix this robustness fact, rather the network of links between pages was designed on purpose in such a way.

6 Limitations of research

While providing a general overview of the inherent structure of Wikipedia links between articles regarding the topic "Squirrel", the exploration presents limitations and space for further analyses. Firstly, the absence of a dictionary to connect pageID (as used in the database) to the actual Wikipedia links or QIDs (Wikipedia contributors n.d.) limits the analysis of results and discussion of them. For instance, it is impossible at the moment to verify whether the article with highest centrality measures was indeed the simple "Squirrel" entry, rather than "Rodent", "Animal", or similar alternatives. Secondly, further analyses could be conducted in comparison with other datasets about different topics of the Wikipedia links database, to show whether the hierarchical structure and small-world characteristics are common to all topics or specific to the explored one. Overall, the exploration gives an overview of how empirical models, specifically related to the information field, do not necessarily follow the same structure as canonical ones, but still show patterns that can be described through comparison with them.

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